

# Lihu Xu

## Applied Probability, Statistics, and Machine Learning

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### Research Profile

Research in applied probability, stochastic analysis, statistics, and machine learning. Current interests include Stein's method and Malliavin calculus; stochastic algorithms (SGD, SVRG, Langevin methods); ergodicity and numerical methods for SDEs/SPDEs; high-dimensional robust inference and time series; and deep-learning-based distribution estimation.

### Academic Appointments

- 2026–present **Professor**, University of Macau, Macau.
- 2026 **Associate Professor (tenured)**, Michigan State University, USA.
- 2019–2026 **Associate Professor**, University of Macau, Macau.
- 2024 **Visiting Faculty**, Harvard University, USA.
- 2014–2019 **Assistant Professor**, University of Macau, Macau.
- 2011–2013 **Lecturer (permanent)**, Brunel University London, UK.
- 2007–2011 Postdoctoral and visiting positions, TU Berlin, Eindhoven University of Technology, and University of Bonn.

### Education

- 2004–2008 Ph.D., Imperial College London, UK.
- 2001–2004 M.S., Peking University, Beijing, China.
- 1997–2001 B.S., Shandong University, Jinan, China.

### Selected Publications in Leading Journals

1. Guillin, A.; Wang, Y.; **Xu, L.**; Yang, H. *Error estimates between SGD with momentum and underdamped Langevin diffusion*. **Annals of Applied Probability**, accepted.
2. Deng, C.; Schilling, R.; **Xu, L.** *Optimal Wasserstein-1 distance between SDEs driven by Brownian motion and stable processes*. **Bernoulli** 31(3), 1834–1857, 2025.
3. Jin, X.; Pang, G.; **Xu, L.**; Xu, X. *An approximation to the invariant measure of the limiting diffusion of  $G/Ph/n + GI$  queues in the Halfin–Whitt regime and related asymptotics*. **Mathematics of Operations Research** (2), 783–812, 2025.
4. Chen, P.; Deng, C.; Schilling, R.; **Xu, L.** *Approximation of the invariant measure of stable SDEs by an Euler–Maruyama scheme*. **Stochastic Processes and Their Applications** 163, 136–167, 2023.
5. **Xu, L.**; Yao, F.; Yao, Q.; Zhang, H. *Non-asymptotic guarantees for robust statistical learning under infinite second moment assumption*. **Journal of Machine Learning Research** 24, 1–46, 2023.
6. Chen, P.; Shao, Q.-M.; **Xu, L.** *A probability approximation framework: Markov process approach*. **Annals of Applied Probability** 33(2), 1619–1659, 2023.
7. Deng, C.; Schilling, R.; **Xu, L.** *Singular integrals of subordinators with applications to structure properties of SPDEs*. **Transactions of the American Mathematical Society** 375, 6043–6073, 2022.
8. Lu, J.; Tan, Y.; **Xu, L.** *Central limit theorem and self-normalized Cramer type moderate deviation for Euler–Maruyama scheme*. **Bernoulli** 28(2), 937–964, 2022.
9. Chen, P.; Jin, X.; Li, X.; **Xu, L.** *A generalized Catoni's  $M$ -estimator under finite  $\alpha$ -th moment assumption with  $\alpha \in (1, 2)$* . **Electronic Journal of Statistics** 15(2), 5523–5544, 2021.
10. Shen, H.; Song, J.; Sun, R.; **Xu, L.** *Scaling limit of a directed polymer among a Poisson field of independent walks*. **Journal of Functional Analysis** 281(5), 109066, 2021.
11. Jia, J.; Xie, F.; **Xu, L.** *Sparse Poisson regression with penalized weighted score function*. **Electronic Journal of Statistics** 13(2), 2898–2920, 2019.
12. **Xu, L.** *Approximation of stable law in Wasserstein-1 distance by Stein's method*. **Annals of Applied Probability** 29(1), 458–504, 2019.
13. Fang, X.; Shao, Q.-M.; **Xu, L.** *Multivariate approximations in Wasserstein distance by Stein's method and Bismut's formula*. **Probability Theory and Related Fields** 174(3–4), 945–979, 2019.
14. Sun, X.; Xiao, Y.; **Xu, L.**; Zhai, J. *Uniform dimension results for a family of Markov processes*. **Bernoulli** 24(4B), 3924–3951, 2018.
15. Wang, R.; **Xu, L.** *Asymptotics for stochastic reaction-diffusion equation driven by subordinate Brownian motion*. **Stochastic Processes and Their Applications** 128(5), 1772–1796, 2018.
16. Wang, R.; Xiong, J.; **Xu, L.** *Irreducibility of stochastic real Ginzburg–Landau equation driven by alpha-stable noises and applications*. **Bernoulli** 23(2), 1179–1201, 2017.
17. Chen, X.; Shao, Q.-M.; Wu, W. B.; **Xu, L.** *Self-normalized Cramer-type moderate deviations under dependence*. **Annals of Statistics** 44(4), 1593–1617, 2016.
18. Wang, F.-Y.; **Xu, L.**; Zhang, X. *Gradient estimates for SDEs driven by multiplicative Lévy noise*. **Journal of Functional Analysis** 269(10), 3195–3219, 2015.
19. **Xu, L.** *Ergodicity of the stochastic real Ginzburg–Landau equation driven by alpha-stable noises*. **Stochastic Processes and Their Applications** 123(10), 3710–3736, 2013.
20. Priola, E.; Shirikyan, A.; **Xu, L.**; Zabczyk, J. *Exponential ergodicity and regularity for equations with Lévy noise*. **Stochastic**

- Processes and Their Applications 122(1), 106–133, 2012.
21. Romito, M.; **Xu, L.** *Ergodicity of the 3D stochastic Navier–Stokes equations driven by mildly degenerate noise.* **Stochastic Processes and Their Applications** 121(4), 673–700, 2011.
  22. **Xu, L.**; Zegarliński, B. *Existence and exponential mixing of infinite white alpha-stable systems with unbounded interactions.* **Electronic Journal of Probability** 15, 1994–2018, 2010.

### Recent Research Funding (Principal Investigator)

- **FDCT/0074/2023/R1A2**, “Applications of methods in stochastic analysis to stochastic algorithms in data science,” 2024–2026, MOP 1,264,000.
- **NSFC 12071499**, “Approximation to ergodic distributions of stochastic (partial) differential equations and applications,” 2021–2024, RMB 510,000.
- **FDCT/0090/2019/A2**, “Applications of Stein’s method in high-dimensional probability approximations,” 2020–2022, MOP 1,237,000.
- **NSFC 11571390**, “Ergodicity of stochastic PDEs driven by highly degenerate pure jump noises,” 2016–2019, RMB 520,000.
- PI on seven University of Macau research grants (2013–2025), including grants on DNN-based GANs, queueing models, high-dimensional probability, SPDEs, and LASSO; total listed funding exceeds MOP 2.9M.

### Supervision and Mentoring

- Supervised **9 graduated Ph.D. students**, **1 co-supervised Ph.D. student**, and **10 master’s students**; currently supervises **3 Ph.D. students**. Mentored **1 postdoctoral researcher**.
- Supervised **16 undergraduate researchers**, plus honor-project, summer-research, visiting Ph.D., and college students. Former students hold academic positions at, among others, Beihang University, Nanjing University of Aeronautics and Astronautics, University of Essex, and Yunnan Normal University.

### Teaching

**University of Macau:** Reinforcement Learning; Bayesian Inference and MCMC; Computational Statistics; Data-Driven Sampling Methods; Applied Statistics; Stochastic Calculus; Probability; High-Dimensional Probability; Stochastic Differential Equations; Advanced Probability.

**Michigan State University:** Probability. **Brunel University London / TU Berlin:** Probability Theory; Mathematical Analysis; Financial Mathematics; Stochastic PDEs.

### Honors, Leadership, and Professional Service

- **Teaching Excellence Award**, Faculty of Science and Technology, University of Macau (2018/19; 2023/24); **Research Excellence Award** (2016/17).
- Associate Editor, *Journal of Theoretical Probability* (2026–present). Invited lecturer and Junior Scientific Leader, Banach International Mathematical Center (2018).
- PI, Mathematics of Machine Learning Section, Center for Applied Mathematics, UM (2020–present); leader, Applied Probability research group (2018–present); chair, Minor Program in Mathematics (2016–present).
- Organizer of probability/statistics seminars and international conference sessions; reviewer for 60+ journal papers and grant applications including ERC, EPSRC, NSFC, and Czech Science Foundation.

### Selected Invited Talks and Skills

Invited talks include University of Minnesota Probability Webinar (2026), IMS Asia Pacific Rim Meeting (2026), HKSIAM Biennial Conference (2025), ICSA China Conference (2025), and multiple international workshops on stochastic analysis, probability approximation, and robust statistics.

**Languages:** English, Chinese. **Computing:** Python, MATLAB, C.